

HEL, Poland

WMO: 121350

Lat: 54.6036N

Lon: 18.8119E

Elev: 3

StdP: 101.29

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.1	-7.0	-12.2	1.3	-8.6	-10.0	1.6	-6.1	13.0	4.8	11.2	3.9	3.9	180	0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.6	25.6	20.4	24.1	19.5	22.6	18.7	21.2	24.4	20.2	23.1	19.3	21.8	3.5	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.1	14.8	23.3	19.1	13.9	22.0	18.2	13.1	21.0	61.7	24.5	58.1	23.2	55.1	21.9	25.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.5	8.3	7.3	-10.8	28.6	3.0	2.1	-13.0	30.1	-14.7	31.3	-16.4	32.5	-18.6	34.0		
			-11.4	22.9	2.9	1.1	-13.4	23.8	-15.1	24.4	-16.8	25.1	-18.8	25.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	0.1	0.5	2.6	6.8	11.5	15.5	18.2	18.4	14.7	9.7	5.4	2.1
	DBStd	7.19	3.81	3.57	2.99	3.23	3.36	2.68	2.41	2.24	2.43	2.90	2.95	3.61
	HDD10.0	1355	306	266	228	106	21	0	0	0	1	41	139	246
	HDD18.3	3536	564	499	486	346	214	94	33	27	112	267	388	504
	CDD10.0	927	0	0	0	10	67	164	254	259	141	33	1	0
	CDD18.3	66	0	0	0	0	1	7	28	28	2	0	0	0
	CDH23.3	213	0	0	0	1	6	33	102	68	3	0	0	0
	CDH26.7	23	0	0	0	0	0	4	14	5	0	0	0	0
Wind	WSAvg	3.9	4.6	4.3	4.0	3.6	3.4	3.5	3.5	3.5	3.7	4.2	4.4	4.7
Precipitation	PrecAvg	580	34	28	31	31	51	57	79	67	61	53	48	41
	PrecMax	715	76	51	55	71	92	128	168	150	144	121	103	69
	PrecMin	417	6	2	6	2	18	6	11	6	15	12	11	18
	PrecStd	76	17	12	13	17	23	28	35	37	33	28	23	14
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.5	8.1	11.7	19.7	24.0	26.9	28.5	27.2	23.3	17.2	12.3	9.8
		MCWB	6.9	6.6	8.1	13.6	18.3	20.5	21.9	21.4	18.6	15.0	11.0	8.6
	2%	DB	6.9	6.9	9.3	16.3	21.3	23.9	25.9	25.1	20.8	15.8	10.9	8.5
		MCWB	5.7	5.6	6.6	11.7	16.5	18.7	20.7	20.3	17.4	13.8	9.8	7.4
	5%	DB	5.9	5.9	7.8	13.9	19.0	21.8	24.2	23.7	19.4	14.8	10.0	7.5
		MCWB	4.8	4.8	5.8	10.2	14.9	17.5	19.8	19.5	16.4	13.2	9.0	6.6
	10%	DB	4.8	5.1	6.6	11.9	17.0	20.2	22.6	22.4	18.3	13.8	9.2	6.5
		MCWB	4.0	4.1	5.1	9.0	13.8	16.5	19.0	18.8	15.7	12.4	8.2	5.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.4	7.1	8.9	14.6	18.7	21.2	22.9	22.4	19.4	15.5	11.4	8.8
		MCDB	8.4	7.8	10.8	18.9	22.8	25.7	26.7	25.9	22.1	16.8	11.9	9.5
	2%	WB	6.0	5.9	7.1	12.2	17.0	19.4	21.6	21.0	17.9	14.3	10.0	7.8
		MCDB	6.7	6.6	8.7	15.6	20.3	22.8	25.0	24.0	20.0	15.4	10.7	8.4
	5%	WB	4.9	4.9	6.1	10.6	15.5	18.1	20.6	20.1	17.0	13.4	9.1	6.7
		MCDB	5.7	5.7	7.5	13.5	18.4	21.0	23.3	22.8	18.9	14.5	9.9	7.3
	10%	WB	4.0	4.1	5.2	9.3	14.0	16.9	19.5	19.3	16.1	12.6	8.3	5.7
		MCDB	4.7	5.0	6.4	11.5	16.7	19.5	22.0	21.6	17.8	13.7	9.1	6.5
Mean Daily Temperature Range	5% DB	MDBR	2.8	3.4	4.9	7.3	7.9	7.3	6.8	6.6	5.2	4.1	3.0	2.8
		MCDBR	3.2	3.9	6.9	10.7	10.6	9.9	9.3	8.4	6.7	5.0	3.5	3.2
		MCWBR	2.9	3.3	4.7	6.6	6.1	5.6	4.9	4.5	4.2	3.8	3.2	3.1
	5% WB	MCDBR	3.2	3.9	6.4	9.9	10.1	9.3	8.5	7.8	6.4	4.8	3.4	3.2
		MCWBR	3.0	3.4	4.7	6.5	6.1	5.8	4.8	4.6	4.2	3.8	3.2	3.2
Clear-Sky Solar Irradiance	taub	0.313	0.341	0.366	0.409	0.399	0.396	0.427	0.419	0.379	0.361	0.346	0.312	
	taud	2.298	2.305	2.298	2.231	2.333	2.376	2.306	2.329	2.418	2.437	2.329	2.294	
	Ebn at Noon	622	734	802	813	846	852	814	797	785	709	574	537	
	Edh at Noon	57	80	100	122	117	113	120	110	89	71	57	48	
All-Sky Solar Radiation	RadAvg	0.46	1.12	2.39	4.22	5.50	5.77	5.17	4.41	3.11	1.56	0.59	0.33	
	RadStd	0.06	0.17	0.27	0.51	0.53	0.47	0.57	0.43	0.34	0.24	0.07	0.04	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	-124	+74	N/A
Regional (22 neighbors)		+0.40	N/A	N/A	N/A	N/A	N/A	N/A	-135	+66	N/A

Nomenclature: See separate page